

QU, Li-yang

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Research Interests: InSAR Time Series Analysis, Coastal Geodesy.

EDUCATION

XIANGTAN UNIVERSITY

Xiangtan, CN

M.S. in Civil and Hydraulic Engineering, Concentrations in Geodesy Sep. 2021- Jun. 2024

- GPA: **3.72/4.0** (Ranking 1/128)
- Awards: **Xiangtan University Academic Scholarship (2021-2023)**; Outstanding Graduate Student Leader (2022); Advanced Individual of Volunteering (2022)

XIANGTAN UNIVERSITY

Xiangtan, CN

B.S. in Surveying and Mapping Engineering

Sep. 2017- Jun. 2021

RESEARCH EXPERIENCE

Research on Key Technologies for Settlement Monitoring and Early Warning of Super High rise Buildings Based on GB-SAR (No. 2B0950)

Participant

Hunan Provincial Department of Education Youth Project

Dec. 2022- Jun. 2025

- Replaced the moving Least Squares method with the Total Least Squares method for elevation fitting and achieved a 12% improvement in fitting accuracy (Working Paper).

Research on the Inversion Method of Spatial Distribution Characteristics of Goaf Based on D-InSAR Technology (No. 2022JJ60014)

Participant

Hunan Provincial Natural Science Foundation Project

Oct. 2022- Jun. 2024

- Processed **87** Sentinel-1A SLC datasets using MT-InSAR techniques to derive 3D surface deformation fields in the Xiangtan manganese mining area. Proved the **lagged correlation (~1 month)** between precipitation and surface subsidence rates by using Spearman correlation analysis. Evaluated local remediation projects using MT-InSAR time series analysis.

Image Exploration of the Effect of Water Mist and Dust Concentration on Electromagnetic Wave Ranging Capability (No.2019RS1059)

Participant

Hunan High level Talent Gathering Project Innovation Team

Apr. 2022- Sep. 2022

- Collected 21 sets of total station ranging data under controlled environmental conditions. Proposed An Improved Simple Adjustment Method for Single Directional Connecting Traverse, increasing fitting accuracy by **40.48%–63.13%**. (Working Paper).

Geological Hazard Risk Survey in Pingjiang County

Participant

Horizontal Research Project

Sep. 2021- Mar. 2022

- Led a team of 5 to produce 1:5000 hazard susceptibility maps using ArcGIS. Applied spatial statistics for regional vulnerability assessment, passing expert peer review.

PATANTS

- WU J, YAO X, LIU Z, HE Y, QU L, XIE Y. Support device based on synthetic aperture radar operational measurement [P]. Hunan Province: CN221629177U, 2024-08-30.

- **QU L**, YAO X, LIU Z, XIE Y, WU J. A synthetic aperture radar interferometry adjustable platform [P]. Hunan Province: CN221463308U, 2024-08-02.
- XIE Y, YAO X, LIU Z, WANG Q, **QU L**, WU J. A protective total station for field geodetic data measurement [P]. Hunan Province: CN221376659U, 2024-07-19.

PUBLICATIONS

- **QU L**. Surface Deformation Monitoring and Analysis of Xiangtan Manganese Mine Based on SBAS-InSAR (in Chinese)[D]. Xiangtan University, 2024.
DOI:10.27426/d.cnki.gxtdu.2024.002039.

WORK AND INTERSHIP EXPERIENCE

CSCEC

Shenzhen, CN

Land Surveyor

Jul. 2024- Present

- Managed the full-cycle survey of construction projects. Assessed the feasibility of applying time-series InSAR technology for predicting structural deformation in ongoing constructions. Benchmarked the performance of multi-source altimetry (**GNSS/InSAR**) in coastal environments, addressing challenges from **atmospheric decorrelation** and ocean **tide loading effects**.

QINGDAO METRO

Qingdao, CN

Land Surveyor (Metro 15# Monitoring Supervisor)

Oct. 2023- Jun. 2024

- Led the team to collect geological, hydrological, and structural deformation monitoring data during subway construction; Explored InSAR research in the field of online engineering; Wrote **334** monitoring reports.

XIANGTAN SURVEY AND DESIGN INSTITUTE

Xiangtan, CN

Assistant Land Surveyor

Jun. 2021- Sep. 2023

- Conducted underground pipeline detection and topographic mapping for **178** urban plots. Performed land ownership registration and cadastral mapping for **107** rural plots.

TECHING EXPERIENCE

XIANGTAN UNIVERSITY

Xiangtan, CN

Teaching Assistant

Sep. 2021- Jan. 2023

- Served as a teaching assistant for a total of **8** undergraduate courses, including photogrammetry, satellite positioning principles and applications, and C# surveying program design, etc. Checked **over 200** assignments and guided undergraduate students in **6** course internships, the assessment results were all "**Excellent**".

SKILLS AND CERTIFICATES

- **Computer Skills:** MS Office, ENVI; SARscape; ArcGIS pro; PyCharm.
- **Certificates:** NCRE-2(Python), Driving License (Full Classes:1 3 22).
- **Languages:** Mandarin (Native), English (CET-6:432).
- **Interests:** Technical Blogger ([See here](#), number of fans:19685).